

SmartReceipt

Project Overview

SmartReceipt is an intelligent mobile application designed to automatically store, analyze, and manage receipts and invoices. Its core mission is to simplify personal finance management by leveraging optical character recognition (OCR) and artificial intelligence (AI) to extract and interpret key data from photographed receipts.

Beyond basic organization, SmartReceipt acts as a personal financial assistant — offering insights, predictions, and visual analytics to help users understand and improve their spending habits. The app stores all data locally for privacy, with optional export features for reporting and synchronization.

Objectives

- Automate the process of collecting and organizing receipts
- Use OCR and AI to extract and interpret key financial information
- Provide intelligent insights and recommendations based on spending behavior
- Offer data visualization and advanced analytics for better financial awareness
- Allow users to share expenses, generate reports, and set reminders for bills

Application Architecture

SmartReceipt follows a modular architecture designed for scalability and maintainability:

- UI Layer— Flutter-based screens for receipt capture, viewing, and analytics
- Data Layer— SQLite database for secure local storage
- Service Layer— Handles OCR, AI analysis, notifications, and report generation
- Model Layer— Defines entities such as Receipt, Category, Group, and Reminder
- State Management— Implemented via MVVM or Provider for scalability and clarity

Main Features

Receipt Scanning and Recognition

Users can capture receipts directly within the app. Using Google ML Kit's Text Recognition, SmartReceipt extracts information such as the store name, total amount, date, and category. AI algorithms then refine this data for accuracy and categorization.

AI Spending Coach

A built-in AI assistant analyzes user spending patterns and offers personalized suggestions to encourage better budgeting. For example, it can detect recurring spending habits and provide friendly, contextual advice:

This feature transforms SmartReceipt from a simple tracker into an interactive financial advisor.

Smart Receipt Summarizer

AI automatically generates summaries of monthly expenses in natural language, highlighting top categories, vendors, and trends:

Users can also opt to receive these summaries as voice reports.

Visual Spending Map

SmartReceipt uses receipt metadata and optional GPS input to create an interactive spending map. Users can visualize where they spend the most through location-based heatmaps:

This feature helps users recognize location-based spending patterns.

Smart Groups

SmartReceipt supports shared expense groups for friends, families, or teams. Members can contribute receipts to a shared space, and the app automatically splits and tracks group expenses, ensuring transparency and fairness.

Local Data Storage

All extracted data is securely stored locally using a lightweight SQLite database, ensuring offline functionality and privacy. Each record includes detailed transaction information and the corresponding receipt image.

Data Viewing and Filtering

Users can view and filter receipts by date, category, or store name. The home screen also displays monthly spending totals and trends.

Statistics and Reports

A dedicated dashboard provides visual charts for easy analysis of spending trends by month or category. Users can export their data as PDF or Excel reports for budgeting, tax, or record-keeping purposes.

Bill Reminders

Users can create reminders for upcoming bills or recurring payments. SmartReceipt sends local notifications before the due date to help avoid missed payments.

Technology Stack

User Interface Overview

- Home Screen— Displays recent receipts, total monthly spending, and shortcuts to add new receipts or access AI insights
- Add Receipt Screen— Capture or upload a receipt, verify extracted data, and assign a category

- Receipt Details Screen— View complete details of a transaction with edit options
- Statistics Screen— Visualize spending trends with bar, pie, and map-based charts
- Smart Coach Screen— Interact with the AI Spending Coach and review personalized suggestions
- Group Expenses Screen— Manage shared groups and view automated expense splits
- Settings Screen— Configure export options, reminders, and privacy preferences

Future Development

- Cloud synchronization for multi-device access
- AI-based item-level recognition for productized receipt parsing
- Multi-language support and smart translation of foreign receipts
- Dark mode and customizable dashboard themes
- Integration with financial APIs or bank SMS import for automatic tracking